NATHANYL L. FREESE

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EDUCATION

Embry-Riddle Aeronautical University (ERAU) – Daytona Beach Campus **Exp. Grad:** December 2026
Bachelor of Science, Aerospace Engineering **GPA:** 3.97/4.00
Tau Beta Pi Engineering Honor Society

WORK EXPERIENCE

Collins Aerospace Internship, Melbourne, FL May 2026 – Present
System Engineering Intern

- Developing Python automation simulating keyboard/mouse input to run 301 transponder test files, cutting test runtime by 50%
- Evaluating an internal AI test case generation tool across 50+ requirements, implementing feedback to improve accuracy
- Developing formal verification procedures while tracking requirements and version control using Jama, JIRA, and GitLab

KBR & LZ Technology Internship, NASA Johnson Space Center, TX January 2025 – August 2025
Engineering Intern

- Developed software tools using Python and PyQt5 to visualize antenna blockage, tracking, and Orion/Starliner simulations with live telemetry overlays for controller training and simulated missions
- Optimized the Gateway visualization tool achieving a 13x overall performance boost and 21x faster graphics rendering
- Led development of the Starliner visualization tool, delivering integrated telemetry overlays used in operational workflows

Thermodynamics Tutor, ERAU, FL January 2024 – January 2025

- Tutored over 400 students seeking help in Thermodynamics and Heat Transfer each semester
- Created and provided targeted problem sets, exam reviews, and custom lesson plans to boost students' understanding

PROJECT EXPERIENCE

Mayott Aerospace Heavy Lift UAVs, Daytona Beach, FL April 2026 – Present
CAD & Simulation Engineer

- Modeling heavy-lift UAV frame components and a 300+ lb payload attachment system in Fusion 360 and SolidWorks, optimizing for strength-to-weight ratio and manufacturability
- Developing a photorealistic Unreal Engine 5 simulation integrated with PX4 to support simulation driven testing and validation of autonomous flight capabilities

Crewed Lunar Mission Preliminary Design, ERAU, FL January 2026 – May 2026
Project & Systems Lead

- Led a 6-person team to design a crewed two-stage LOX/CH4 lunar vehicle covering launch through splashdown
- Performed vehicle sizing in MATLAB and CEA and led trade studies across all subsystems to meet mission requirements
- Modeled full launch vehicle assembly in CATIA V.5 and Fusion 360, producing detailed parts breakdown

Embry-Riddle Future Space Explorers and Development (ERFSEDS) Artemini Rocket, ERAU, FL August 2024 – January 2025
Manufacturing Member

- Modeled a scaled down Artemis rocket in CATIA V.5 and SolidWorks, using OpenRocket to optimize flight characteristics
- Collaborated with a 15-member team to integrate design modifications and ensure structural reliability

Experimental Rocket Propulsion Lab (ERPL) Spectre Rocket, ERAU, FL August 2022 – May 2024
Hardware Member

- Contributed to the design, assembly, and testing of an active stabilization system, including canard integration, for a high-powered rocket within a 10-member engineering team
- Modeled structural components in Fusion 360, soldered avionics hardware, and analyzed test data in MATLAB

SKILLS

- CATIA V.5	- SolidWorks	- Fusion 360	- Creo Parametric	- NASTRAN	- Ansys Fluent
- MATLAB	- Python	- Unreal Engine	- Programming C	- PyQt5	- GitLab / GitHub
- VS Code	- JIRA	- JAMA	- NASA CEA	- Microsoft Office	- Linux CLI